

Finite Element Method

Lecture: Hyperelastic Models, Stress Tensors,
Weak Form and Linearisation
MECH4103 — Spring 2026

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- 1 Stress Tensors and Work Conjugacy
- 2 Hyperelastic Constitutive Theory
- 3 Embedded Exercises on Constitutive Models
- 4 Weak Form of Equilibrium
- 5 Linearisation for Newton's Method
- 6 Embedded Exercises on Linearisation
- 7 Summary and Outlook

Cauchy stress σ

- True stress in the **current** configuration.
- Cauchy's theorem: $\mathbf{t} = \sigma \mathbf{n}$.
- **Symmetric**: $\sigma = \sigma^T$.

First Piola–Kirchhoff stress $\mathbf{P}$

$$\mathbf{P} = J\sigma\mathbf{F}^{-T}$$

- Force in current config. per **reference area**.
- Two-field tensor (one leg in each config.).
- **Non-symmetric** in general.

Second Piola–Kirchhoff stress $\mathbf{S}$

$$\mathbf{S} = \mathbf{F}^{-1}\mathbf{P} = J\mathbf{F}^{-1}\sigma\mathbf{F}^{-T}$$

- Fully referred to **reference** configuration.
- **Symmetric**: $\mathbf{S} = \mathbf{S}^T$.
- Work-conjugate to $\mathbf{E}$: $\mathbf{S} \cdot \dot{\mathbf{E}}$.
- Central quantity in hyperelastic constitutive theory.

Kirchhoff stress τ

$$\tau = J\sigma = \mathbf{F}\mathbf{S}\mathbf{F}^T$$

- Push forward of $\mathbf{S}$ to current config.
- Convenient for spatial formulations.
- Work-conjugate to $\mathbf{d}$.

Tensor	Config.	Symmetric
σ	current	✓
$\tau = J\sigma$	current	✓
$\mathbf{P}$	mixed	×
$\mathbf{S}$	reference	✓

Work conjugate pairs

$$\mathbf{S} \cdot \dot{\mathbf{E}} = \tau \cdot \mathbf{d} = \frac{1}{2} \mathbf{S} \cdot \dot{\mathbf{C}}$$

Objectivity note: $\mathbf{S}$ and $\mathbf{E}$ live in the fixed reference configuration — they are *always* objective and never contaminated by observer spin. No special treatment needed. By contrast, rate laws in the current config. (e.g.

Strain energy function $W(\mathbf{C})$

A material is **hyperelastic** if there exists a strain energy function W (per unit reference volume) such that:

$$\boxed{\mathbf{S} = 2 \frac{\partial W}{\partial \mathbf{C}}}, \quad S_{AB} = 2 \frac{\partial W}{\partial C_{AB}}$$

- The Cauchy stress follows via push forward:
 $\boldsymbol{\sigma} = J^{-1} \mathbf{F} \mathbf{S} \mathbf{F}^T = 2\rho \mathbf{b} \partial W / \partial \mathbf{b}$.
- For **isotropic** materials W depends only on the **invariants** of $\mathbf{C}$ (or equivalently of $\mathbf{b}$):

$$W = W(I_C, II_C, III_C)$$

- Invariants: $I_C = \text{tr } \mathbf{C}$, $II_C = \frac{1}{2}[I_C^2 - \text{tr}(\mathbf{C}^2)]$,
 $III_C = \det \mathbf{C} = J^2$.
- Or equivalently in principal stretches: $W = W(\lambda_1^2, \lambda_2^2, \lambda_3^2)$.

Why hyperelasticity?

- Stress derived from a **potential** $\Rightarrow$ automatic thermodynamic consistency.
- **No time integration** needed for the constitutive equation (unlike plasticity/viscoplasticity).
- Straightforward **linearisation**: tangent stiffness from second derivative of W .
- Suitable for rubber, foams, biological tissue, polymer nanocomposites at moderate rates.

Thermodynamic consistency

The Clausius–Duhem inequality reduces to $\mathcal{D} = \mathbf{S} \cdot \dot{\mathbf{E}} - \dot{W} = 0$ for purely elastic response — identically satisfied by construction.

Neo-Hooke material

$$W = \underbrace{U(J)}_{\text{volumetric}} + \underbrace{\frac{1}{2}\mu(I_C - 3)}_{\text{isochoric}}$$

With $U(J) = \frac{\Lambda}{4}(J^2 - 1) - \frac{\Lambda}{2} \ln J$:

$$\mathbf{S} = \frac{\Lambda}{2}(J^2 - 1)\mathbf{C}^{-1} + \mu(\mathbf{1} - \mathbf{C}^{-1})$$

Cauchy stress:

$$\boldsymbol{\sigma} = \frac{\Lambda}{2J}(J^2 - 1)\mathbf{1} + \frac{\mu}{J}(\mathbf{b} - \mathbf{1})$$

- Only two parameters: Λ , μ (Lamé constants).
- Reduces to Hooke's law for small strains.
- Satisfies growth conditions: $W \rightarrow \infty$ as $J \rightarrow 0^+$ or $J \rightarrow +\infty$.

Mooney–Rivlin material

$$W = c_1(I_C - 3) + c_2(II_C - 3)$$

- Two parameters c_1, c_2 (incompressible).
- Obtained from Ogden model with $r = 2$, $\alpha_1 = 2$, $\alpha_2 = -2$.
- Better fit to rubber data than Neo-Hooke for moderate stretches.
- Reduction: $c_2 = 0 \Rightarrow$ Neo-Hooke with $\mu = 2c_1$.

Polyconvexity condition (existence of solutions)

For Ogden-type models: $\sum_i \mu_i \alpha_i = 2\mu$ and $\mu_i \alpha_i > 0$. Ensures linearisation yields positive-definite tangent stiffness near $\mathbf{F} = \mathbf{1}$.

Ogden strain energy (compressible)

$$W(\lambda_k) = \sum_{i=1}^r \frac{\mu_i}{\alpha_i} (\lambda_1^{\alpha_i} + \lambda_2^{\alpha_i} + \lambda_3^{\alpha_i} - 3) + g(J)$$

- Parameters: μ_i , α_i (real); $g(J)$ volumetric term.
- Three-term version ($r = 3$) fits rubber data up to **700% strain** (Treloar 1944 experiments).
- 2nd Piola–Kirchhoff stress in principal directions:

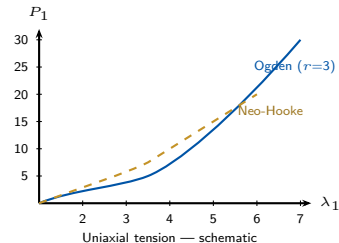
$$S_{(i)} = \frac{1}{\lambda_{(i)}} \frac{\partial W}{\partial \lambda_{(i)}}$$

- Kirchhoff stress: $\tau_{(i)} = \lambda_{(i)} \partial W / \partial \lambda_{(i)}$

Ogden parameters (Treloar data)

i	μ_i	α_i
1	6.30	1.3
2	0.013	5.0
3	-0.10	-2.0

Check: $\sum \mu_i \alpha_i = 8.19 + 0.065 - 0.20 \approx 2\mu$



- For rubber-like and foam materials, the energy is split:

$$W(\mathbf{C}, J) = \hat{W}(\bar{\mathbf{C}}) + U(J)$$

with $\bar{\mathbf{C}} = J^{-2/3} \mathbf{C}$ (isochoric part of $\mathbf{C}$).

- The 2nd Piola–Kirchhoff stress splits accordingly:

$$\mathbf{S} = \underbrace{\mathbf{S}^{\text{iso}}}_{\text{shape}} + \underbrace{\mathbf{S}^{\text{vol}}}_{\text{volume}}$$

$$\mathbf{S}^{\text{vol}} = J \frac{\partial U}{\partial J} \mathbf{C}^{-1}, \quad \mathbf{S}^{\text{iso}} = \mathbb{P} \left(2 \frac{\partial \hat{W}}{\partial \bar{\mathbf{C}}} \right)$$

where $\mathbb{P} = \mathbb{I} - \frac{1}{3} \mathbf{C}^{-1} \otimes \mathbf{C}$ is the isochoric projection.

- Kirchhoff stress split:

$$\boldsymbol{\tau} = \underbrace{Jp \mathbf{1}}_{\boldsymbol{\tau}^{\text{vol}}} + \underbrace{\text{dev } \bar{\boldsymbol{\tau}}}_{\boldsymbol{\tau}^{\text{iso}}}, \quad p = \frac{\partial U}{\partial J}$$

Neo-Hooke: explicit split

$$\mathbf{S}^{\text{iso}} = \mu J^{-2/3} \left(\mathbf{1} - \frac{1}{3} \text{tr } \mathbf{C} \mathbf{C}^{-1} \right)$$

$$\mathbf{S}^{\text{vol}} = \frac{K}{2} (J^2 - 1) \mathbf{C}^{-1}$$

with $K = \Lambda + \frac{2}{3} \mu$ (bulk modulus).

Relevance for FEM

The split is **essential** for nearly incompressible problems:

- Pressure p and volumetric dilation $\theta \approx J$ treated as **independent fields**.
- Leads to the **three-field Hu–Washizu** functional $\Pi(\varphi, p, \theta)$ and $\bar{B}$ elements that avoid volumetric locking.

Exercise

Starting from the Neo-Hooke strain energy $W(I_C, J) = U(J) + \frac{1}{2}\mu(I_C - 3)$, express the **Cauchy stress tensor** $\boldsymbol{\sigma}$ in terms of the left Cauchy–Green tensor $\mathbf{b} = \mathbf{F}\mathbf{F}^\top$.

Solution

Starting from $\boldsymbol{\sigma} = J^{-1}\mathbf{F}\mathbf{S}\mathbf{F}^\top$ with $\mathbf{S} = 2\partial W/\partial \mathbf{C}$:

$$\boldsymbol{\sigma} = 2\rho \mathbf{b} \frac{\partial W}{\partial \mathbf{b}} = \frac{2\rho}{1} \left[\left(\frac{\partial W}{\partial I_b} + I_b \frac{\partial W}{\partial II_b} \right) \mathbf{b} - \frac{\partial W}{\partial II_b} \mathbf{b}^2 + III_b \frac{\partial W}{\partial III_b} \mathbf{1} \right]$$

For the Neo-Hooke model $W = U(J) + \frac{\mu}{2}(I_b - 3)$, and using $\mathbf{F}\mathbf{C}^{-1}\mathbf{F}^\top = \mathbf{1}$:

$$\boldsymbol{\sigma} = \frac{\Lambda}{2J}(J^2 - 1)\mathbf{1} + \frac{\mu}{J}(\mathbf{b} - \mathbf{1})$$

At the undeformed state $J = 1$, $\mathbf{b} = \mathbf{1}$: $\boldsymbol{\sigma} = \mathbf{0}$. ✓

As $J \rightarrow 0$: $\boldsymbol{\sigma} \rightarrow -\infty$ (correct physical limit). ✓

Exercise

A rod is stretched uniaxially ($\lambda_1 = \lambda$, $\lambda_2 = \lambda_3 = 1$, no lateral contraction assumed). Compare the Cauchy stress σ_1 predicted by **(a)** the Neo-Hooke model (Eq. above) and **(b)** the St. Venant (linear **S–E**) material $\mathbf{S} = \Lambda \operatorname{tr} \mathbf{E} \mathbf{1} + 2\mu \mathbf{E}$. Discuss the limiting behaviour as $\lambda \rightarrow 0$.

Solution

With $J = \lambda_1 = \lambda$, $F_{11} = \lambda$, $b_{11} = \lambda^2$:

(a) Neo-Hooke:

$$\sigma_1^{\text{NH}} = \frac{\Lambda}{2\lambda}(\lambda^2 - 1) + \frac{\mu}{\lambda}(\lambda^2 - 1) = \gamma \left(\lambda - \frac{1}{\lambda} \right), \quad \gamma = \frac{\Lambda}{2} + \mu$$

Limits: $\lambda \rightarrow \infty \Rightarrow \sigma_1 \rightarrow +\infty$; $\lambda \rightarrow 0 \Rightarrow \sigma_1 \rightarrow -\infty$. **Physically correct.**

(b) St. Venant: $E_{11} = \frac{1}{2}(\lambda^2 - 1)$, so $S_1 = \gamma(\lambda^2 - 1)$ and $\sigma_1 = \lambda S_1 = \gamma\lambda(\lambda^2 - 1)$.

As $\lambda \rightarrow 0$: $\sigma_1 \rightarrow 0$. **Physically wrong!** (Infinite compression gives zero stress.)

Conclusion: St. Venant is restricted to small–moderate strains; use hyperelastic models for large-strain simulations.

Strong form (static, reference config.)

$$\text{Div } \mathbf{P} + \rho_0 \bar{\mathbf{b}} = \mathbf{0} \quad \text{in } \mathcal{B}, \quad \mathbf{P}\mathbf{N} = \bar{\mathbf{t}} \quad \text{on } \partial\mathcal{B}_\sigma$$

Weak form (principle of virtual work)

Multiply by test function $\boldsymbol{\eta}$ ($\boldsymbol{\eta} = \mathbf{0}$ on $\partial\mathcal{B}_u$) and integrate by parts:

$$G(\varphi, \boldsymbol{\eta}) = \int_{\mathcal{B}} \mathbf{S} \cdot \delta \mathbf{E} dV - \int_{\mathcal{B}} \rho_0 \bar{\mathbf{b}} \cdot \boldsymbol{\eta} dV - \int_{\partial\mathcal{B}_\sigma} \bar{\mathbf{t}} \cdot \boldsymbol{\eta} dA = 0$$

- $\delta \mathbf{E} = \frac{1}{2}(\mathbf{F}^T \text{Grad } \boldsymbol{\eta} + \text{Grad}^T \boldsymbol{\eta} \mathbf{F})$ — variation of Green–Lagrange strain.
- First term: **internal virtual work**.
- Remaining terms: **external virtual work**.
- No additional assumptions (valid for inelastic materials too, but here $\mathbf{S} = 2\partial W/\partial \mathbf{C}$).

Equivalent spatial form

$$g(\varphi, \boldsymbol{\eta}) = \int_{\varphi(\mathcal{B})} \boldsymbol{\sigma} \cdot \nabla^S \boldsymbol{\eta} dv - \int_{\varphi(\mathcal{B})} \rho \bar{\mathbf{b}} \cdot \boldsymbol{\eta} dv - \int_{\varphi(\partial\mathcal{B}_\sigma)} \hat{\mathbf{t}} \cdot \boldsymbol{\eta} da = 0 \quad (1)$$

where $\nabla^S \boldsymbol{\eta} = \frac{1}{2}(\text{grad } \boldsymbol{\eta} + \text{grad}^T \boldsymbol{\eta})$.

Connection to FEM

The weak form is discretised by:

$$\boldsymbol{\eta} \approx \sum_a N_a \hat{\boldsymbol{\eta}}_a, \quad \varphi \approx \sum_a N_a \hat{\mathbf{x}}_a$$

yielding a system of nonlinear algebraic equations $\mathbf{R}(\mathbf{d}) = \mathbf{0}$ for nodal displacements $\mathbf{d}$.

- For nearly incompressible materials, displacement-only elements **lock** (overly stiff).
- The three-field Hu–Washizu functional introduces pressure p and dilation θ as **independent** variables:

$$\Pi(\varphi, p, \theta) = \int_{\mathcal{B}} \left[\hat{W}(\bar{\mathbf{C}}) + U(\theta) + p(J - \theta) \right] dV - \Pi_{\text{ext}}$$

- Stationarity yields three equations:
 - 1 $\delta\varphi$: equilibrium (standard weak form with split stresses)
 - 2 δp : kinematic constraint $J = \theta$
 - 3 $\delta\theta$: constitutive equation $p = \partial U / \partial \theta$

Spatial form of equation (1)

$$\int_{\mathcal{B}} \left\{ \boldsymbol{\tau}^{\text{iso}} \cdot \nabla^S \boldsymbol{\eta} + \tau^{\text{vol}} \text{Div } \boldsymbol{\eta} - \rho_0 \bar{\mathbf{b}} \cdot \boldsymbol{\eta} \right\} dV \quad (2)$$

$$- \int_{\partial \mathcal{B}_\sigma} \boldsymbol{\eta} \cdot \hat{\mathbf{t}} dA = 0$$

$\bar{\mathbf{B}}$ and mixed elements

Discretisation of the Hu–Washizu principle leads to $\bar{\mathbf{B}}$ -**bar elements** (mean dilatation method) and $Q1P0$ elements — standard tools for rubber-like materials in commercial FEM codes (Abaqus, ANSYS).

Why linearise?

The weak form $G(\varphi, \boldsymbol{\eta}) = 0$ is **nonlinear** in φ . Newton–Raphson iteration requires the tangent:

$$L[G]\bar{\varphi} = G(\bar{\varphi}, \boldsymbol{\eta}) + DG(\bar{\varphi}, \boldsymbol{\eta}) \cdot \Delta \mathbf{u} = 0$$

Directional derivative of G

$$DG(\bar{\varphi}, \boldsymbol{\eta}) \cdot \Delta \mathbf{u} = \int_{\mathcal{B}} \left\{ \underbrace{\text{Grad } \Delta \mathbf{u} \bar{\mathbf{S}} \cdot \text{Grad } \boldsymbol{\eta}}_{\text{initial stress}} + \underbrace{\delta \bar{\mathbf{E}} \cdot \bar{\mathbb{C}}[\Delta \bar{\mathbf{E}}]}_{\text{material}} \right\} dV$$

- **Initial stress term:** accounts for geometric nonlinearity; depends on current stress $\bar{\mathbf{S}}$.
- **Material term:** depends on incremental constitutive tensor $\bar{\mathbb{C}} = 4 \partial^2 W / \partial \mathbf{C} \partial \mathbf{C}$.
- $\Delta \bar{\mathbf{E}} = \frac{1}{2}(\bar{\mathbf{F}}^T \text{Grad } \Delta \mathbf{u} + \text{Grad}^T \Delta \mathbf{u} \bar{\mathbf{F}})$

Spatial (updated Lagrange) form

$$Dg(\bar{\varphi}, \boldsymbol{\eta}) \cdot \Delta \mathbf{u} = \quad (3)$$

$$\int_{\bar{\varphi}(\mathcal{B})} \left\{ \text{grad } \Delta \mathbf{u} \bar{\boldsymbol{\sigma}} \cdot \text{grad } \boldsymbol{\eta} + \nabla^S \boldsymbol{\eta} \cdot {}^c[\nabla^S \Delta \mathbf{u}] \right\} dv$$

with spatial constitutive tensor

$$c_{iklm}^c = J^{-1} F_{iA} F_{lC} F_{mD} F_{kB} C_{ABCD}.$$

The spatial tangent c is the push-forward of the reference tangent $\mathbb{C}$ — this is precisely the push-forward step of the Lie derivative, ensuring the incremental constitutive law is objective.

Newton–Raphson algorithm

Given residual $\mathbf{R}^{(k)}$, solve

$$\underbrace{\mathbf{K}_T^{(k)}}_{\text{tangent stiffness}} \Delta \mathbf{d} = -\mathbf{R}^{(k)}$$

Definition

$$\mathbb{C} = 4 \frac{\partial^2 W}{\partial \mathbf{C} \partial \mathbf{C}}, \quad \dot{\mathbf{S}} = \mathbb{C} \left[\frac{1}{2} \dot{\mathbf{C}} \right]$$

Neo-Hooke (reference config.)

From $\mathbf{S} = \frac{\Lambda}{2} (J^2 - 1) \mathbf{C}^{-1} + \mu (\mathbf{1} - \mathbf{C}^{-1})$:

$$\mathbb{C} = \Lambda J^2 \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + [2\mu - \Lambda(J^2 - 1)] \mathbb{E}^{C^{-1}}$$

where $\mathbb{E}_{ABCD}^{C^{-1}} = -\partial C_{AB}^{-1} / \partial C_{CD} = \frac{1}{2} (C_{AC}^{-1} C_{BD}^{-1} + C_{AD}^{-1} C_{BC}^{-1})$.

At the undeformed state ($\mathbf{F} = \mathbf{1}$, $J = 1$)

$$\mathbb{C}_0 = \Lambda \mathbf{1} \otimes \mathbf{1} + 2\mu \mathbb{E}$$

This is identical to the elasticity tensor of the **linear theory**!

Spatial tangent (current config.)

$$\mathbf{c}^c = \Lambda J^2 \mathbf{1} \otimes \mathbf{1} + [2\mu - \Lambda(J^2 - 1)] \mathbb{E}$$

In Voigt matrix form (6×6):

$$[\mathbf{c}^c] = \begin{bmatrix} 2\mu + \Lambda & \Lambda J^2 & \Lambda J^2 & & & \\ \Lambda J^2 & 2\mu + \Lambda & \Lambda J^2 & & & \\ \Lambda J^2 & \Lambda J^2 & 2\mu + \Lambda & & & \\ & & & \alpha & & \\ & & & & \alpha & \\ & & & & & \alpha \end{bmatrix}$$

$$\alpha = \mu - \frac{\Lambda}{2} (J^2 - 1)$$

The volumetric–isochoric split of $\mathbb{C}$ follows analogously:

$\mathbb{C} = \mathbb{C}^{\text{vol}} + \mathbb{C}^{\text{iso}}$ — see Wriggers (2008),
eq. (3.251)–(3.253).

Exercise

Compute the directional derivative (linearisation) of the Green–Lagrange strain tensor $\mathbf{E} = \frac{1}{2}(\mathbf{F}^T \mathbf{F} - \mathbf{1})$ at a deformation state $\bar{\varphi}$ in the direction of an increment $\Delta \mathbf{u}$. What does it reduce to at the initial state $\varphi = \mathbf{X}$?

Solution

By definition of the directional derivative:

$$\Delta \bar{\mathbf{E}} = D\mathbf{E}(\bar{\varphi}) \cdot \Delta \mathbf{u} = \left. \frac{d}{d\alpha} \right|_{\alpha=0} \frac{1}{2} \left[\mathbf{F}^T(\bar{\varphi} + \alpha \Delta \mathbf{u}) \mathbf{F}(\bar{\varphi} + \alpha \Delta \mathbf{u}) - \mathbf{1} \right]$$

Differentiating and using linearity of Grad:

$$\Delta \bar{\mathbf{E}} = \frac{1}{2} \left(\bar{\mathbf{F}}^T \text{Grad} \Delta \mathbf{u} + \text{Grad}^T \Delta \mathbf{u} \bar{\mathbf{F}} \right)$$

At $\varphi = \mathbf{X}$ (undeformed): $\bar{\mathbf{F}} = \mathbf{1}$, so

$$\Delta \mathbf{E}|_{\varphi=\mathbf{X}} = \frac{1}{2} (\text{Grad} \Delta \mathbf{u} + \text{Grad}^T \Delta \mathbf{u}) = \boldsymbol{\epsilon}(\Delta \mathbf{u})$$

the **linear strain tensor** of classical elasticity. The linearisation of the nonlinear theory *about the reference state* recovers the linear theory.

Exercise

Show that the incremental constitutive tensor $\mathbb{C} = \Lambda J^2 \mathbf{C}^{-1} \otimes \mathbf{C}^{-1} + [2\mu - \Lambda(J^2 - 1)]\mathbb{E}^{C^{-1}}$ for the Neo-Hooke model reduces at $\mathbf{F} = \mathbf{1}$ (i.e. $J = 1$, $\mathbf{C} = \mathbf{1}$) to the classical isotropic elasticity tensor. Write the result in Voigt notation.

Solution

At $\mathbf{F} = \mathbf{1}$: $J = 1$, $\mathbf{C}^{-1} = \mathbf{1}$, $\mathbb{E}^{C^{-1}} \rightarrow \mathbb{E}$ (fourth-order identity).

$$\mathbb{C}_0 = \Lambda \mathbf{1} \otimes \mathbf{1} + 2\mu \mathbb{E}$$

In Voigt notation ($\boldsymbol{\sigma} = \mathbb{C}_0 \boldsymbol{\varepsilon}$):

$$[\mathbb{C}_0] = \begin{bmatrix} 2\mu + \Lambda & \Lambda & \Lambda & & & \\ \Lambda & 2\mu + \Lambda & \Lambda & & & \\ \Lambda & \Lambda & 2\mu + \Lambda & & & \\ & & & \mu & & \\ & & & & \mu & \\ & & & & & \mu \end{bmatrix}$$

This is the standard Lamé elasticity tensor: $E = \mu(3\Lambda + 2\mu)/(\Lambda + \mu)$, $\nu = \Lambda/[2(\Lambda + \mu)]$.

Stress tensors

- $\mathbf{S} = J\mathbf{F}^{-1}\boldsymbol{\sigma}\mathbf{F}^{-\top}$ (2nd PK, reference, symmetric)
- $\boldsymbol{\tau} = J\boldsymbol{\sigma} = \mathbf{F}\mathbf{S}\mathbf{F}^{\top}$ (Kirchhoff, current, symmetric)
- Work pairs: $\mathbf{S}:\dot{\mathbf{E}} = \boldsymbol{\tau}:\mathbf{d}$

Hyperelastic models

- $\mathbf{S} = 2\partial W/\partial \mathbf{C}$ — potential formulation
- Neo-Hooke (Λ, μ) — two-parameter, large strains
- Mooney–Rivlin (c_1, c_2) — better rubber fit
- Ogden (μ_i, α_i) — fits data to 700% strain
- Vol–iso split for (near) incompressibility

Weak form and linearisation

- $G = \int \mathbf{S} \cdot \delta \mathbf{E} dV - W_{\text{ext}} = 0$
- Linearisation $\rightarrow$ tangent stiffness: initial stress + material parts
- $\mathbb{C}_0$ at $\mathbf{F} = \mathbf{1}$ = linear elasticity tensor
- Newton–Raphson: quadratic convergence

What comes next in MECH4103

FEM discretisation of the weak form: isoparametric elements for finite deformations, assembly of $\mathbf{K}_T$, Newton iteration, and implementation in Abaqus UMAT/UHYPER.

Reference

P. Wriggers, *Nonlinear Finite Element Methods*, Springer 2008, Secs. 3.2–3.5.

Thanks for your attention!